

1. SELECT Clause:

1. Write an SQL query to select all columns from the "Employees" table.

```
SELECT * FROM Employee;
```

2. How do you use the SELECT DISTINCT clause to retrieve unique values from a column?

```
SELECT DISTINCT Department  
FROM Employee;
```

3. Write an SQL command to concatenate the "FirstName" and "LastName" columns and alias it as "FullName".

```
SELECT Name AS FullName  
FROM Employee;
```

4. Explain how to use the AS keyword in the SELECT clause.

```
SELECT Salary AS Employee_Salary  
FROM Employee;
```

5. Write an SQL query to calculate the total salary of all employees.

```
SELECT SUM(Salary) AS Total_Salary  
FROM Employee;
```

2. WHERE Clause:

1. Write an SQL query to select employees from the "Employees" table where the "Department" is 'HR'.

```
SELECT *  
FROM Employee  
WHERE Department = 'HR';
```

2. How do you use comparison operators like =, <>, <, >, <=, >= in the WHERE clause?

```
SELECT *
```

```
FROM Employee  
WHERE Salary > 50000;
```

4. Explain how to use logical operators such as AND, OR, and NOT in the WHERE clause.

```
SELECT *  
FROM Employee  
WHERE Department = 'IT'  
AND Salary > 50000;
```

5. Write an SQL query to select orders with a total amount greater than 1000.

```
SELECT *  
FROM Employee  
WHERE Salary > 60000;
```

3. ORDER BY Clause:

2. How do you use the DESC keyword to sort data in descending order?

```
SELECT *  
FROM Employee  
ORDER BY Salary DESC;
```

3. Write an SQL command to sort products based on their salary in descending order and display only the top 3.

```
SELECT *  
FROM Employee  
ORDER BY Salary DESC  
LIMIT 3;
```

4. Explain how to sort records based on multiple columns using the ORDER BY clause.

```
SELECT *  
FROM Employee
```

```
ORDER BY Salary ASC, ID DESC;
```

4. GROUP BY Clause:

1. Write an SQL query to calculate the total number of employees in each department.

```
SELECT Department, COUNT(*) AS Total_Employees  
FROM Employee  
GROUP BY Department;
```

2. How do you use aggregate functions like COUNT(), SUM(), AVG(), MAX(), and MIN() with the GROUP BY clause?

```
SELECT Department,  
       COUNT(*) AS Total_Employees,  
       SUM(Salary) AS Total_Salary,  
       AVG(Salary) AS Average_Salary,  
       MAX(Salary) AS Highest_Salary,  
       MIN(Salary) AS Lowest_Salary  
FROM Employee  
GROUP BY Department;
```

4. Write an SQL query to find the average salary of employees in each department.

```
SELECT Department,  
       AVG(Salary) AS Average_Salary  
FROM Employee  
GROUP BY Department;
```

5. HAVING Clause:

1. Write an SQL query to find departments with more than 5 employees.

```
SELECT Department,  
       COUNT(*) AS Total_Employees  
FROM Employee  
GROUP BY Department  
HAVING COUNT(*) > 5;
```

2. How do you use aggregate functions with the HAVING clause?

```
SELECT Department,  
       AVG(Salary) AS Average_Salary  
FROM Employee  
GROUP BY Department  
HAVING AVG(Salary) > 50000;
```

3. Write an SQL command to retrieve departments with an average salary greater than 50000.

```
SELECT Department,  
       AVG(Salary) AS Average_Salary  
FROM Employee  
GROUP BY Department  
HAVING AVG(Salary) > 50000;
```

4. Write an SQL query to find departments with a total salary greater than 500000.

```
SELECT Department,  
       SUM(Salary) AS Total_Salary  
FROM Employee  
GROUP BY Department  
HAVING SUM(Salary) > 500000;
```

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